

Assessment: Recipe Scraping and Unstructured JSON Management

1. Objective

To evaluate the candidate's ability to:

1. Scrape recipe data from the internet using Python and web scraping libraries.
2. Store and manage unstructured JSON data, demonstrating flexibility in handling inconsistent data schemas.
3. Understand how such unstructured data could later be used by a Large Language Model (LLM) for structured recipe generation and grocery list creation.

2. Problem Description

The candidate will develop a Python-based script or mini-application that:

- Scrapes publicly available recipe information from the internet.
- Extracts relevant details (e.g., ingredients, steps, cooking time, etc.).
- Stores the scraped data as unstructured JSON documents, each representing one recipe.
- Handles variations in structure and missing fields gracefully.

3. Reference Website

Use the following site to understand the general format and content structure of recipes:

<https://recipehubn.netlify.app/> or

You don't need to scrape this site directly — it is provided for conceptual understanding of recipe content (titles, ingredients, steps, etc.).

You may scrape from publicly available recipe blogs or websites that allow educational use.

4. Tasks and Expected Deliverables

Task 1: Recipe Scraping

- Write a Python script that uses BeautifulSoup, Requests, or Scrapy to scrape recipe data.
- Each recipe should attempt to extract:
 - Title
 - Ingredients
 - Quantity (if available)
 - Preparation steps (may be text or structured)
 - Cook/prep time

- Cuisine type (if identifiable)
- Some fields may be missing or formatted differently — handle these cases without breaking the script.

Task 2: JSON Storage

- Store each recipe as a separate unstructured JSON document.
- Example structures:

```
{
  "title": "Chocolate Cake",
  "ingredients": ["Flour - 2 cups", "Cocoa powder - 1 cup", "Eggs - 3"],
  "prep": "Preheat oven to 350°F, mix all ingredients...",
  "steps": ["Mix ingredients", "Bake for 30 minutes"],
  "servings": 6
}
```

```
{
  "RecipeName": "Paneer Butter Masala",
  "Qty": "Serves 4",
  "IngredientsList": ["Paneer cubes", "Tomato puree", "Butter", "Cream"],
  "Instructions": "Cook paneer in butter. Add sauce and simmer."
}
```

- Store all JSON files in a PostgreSQL database that supports JSONB fields.

Task 3: Data Quality and Flexibility

- Demonstrate that your code can handle recipes with missing or extra fields without crashing.
- Show at least 5–10 different recipes, each with unique or inconsistent property names.

5. Stretch Task (Optional, for Extra Credit)

If time permits:

- Write a small Python function that reads these unstructured JSON documents and

normalizes them into a consistent structure (e.g., ensuring every recipe has a title, ingredients, and steps field).

- This simulates the LLM-based structure learning that will be implemented later.

6. Expected Learning Outcomes

By completing this assessment, the candidate will:

- Demonstrate proficiency in Python, web scraping, and JSON handling.
- Learn how to manage inconsistent unstructured data.
- Understand the data pipeline foundation for a larger AI-driven meal planning and grocery ordering system.

7. Evaluation Criteria

Evaluation Area	Description	Weight
Technical Skill (Python)	Efficient use of libraries like BeautifulSoup, Requests, Scrapy	25%
Data Handling	Ability to create and manipulate unstructured JSON documents	25%
Robustness	Error handling, flexibility for missing or inconsistent fields	20%
Code Quality	Readability, comments, logical structure	15%
Documentation & Presentation	Clarity of approach and output examples	15%

8. Submission Requirements

- Source code files (.py scripts or Jupyter notebook).
- A folder or PostgreSQL dump containing all JSON recipe files.
- A brief README.md explaining:
 - The data sources used.
 - How to run the script.
 - Challenges faced and how they were handled.